

FREIBURG, Germany

WMO: 108030

Lat: 48.0233N

Lon: 7.8342E

Elev: 779

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	19.6	7.3	8.2	17.2	12.1	10.5	22.2	29.4	46.7	27.3	48.2	3.9	350	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	88.0	68.1	84.5	66.8	81.4	65.5	69.7	83.7	68.2	81.6	66.9	78.9	8.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	95.5	75.4	63.4	90.1	73.8	62.1	85.7	72.6	34.1	84.1	32.9	81.8	31.8	79.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.2	18.5	DB	13.9	94.6	7.4	2.5	8.5	96.4	4.2	97.9	0.0	99.3	-5.4	101.1	
			WB	12.9	73.3	7.2	1.3	7.7	74.3	3.5	75.0	-0.5	75.8	-5.8	76.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	35.9	36.6	44.7	50.3	58.5	64.8	69.7	69.0	62.4	53.3	43.2	39.1
	DBStd	14.05	9.96	9.33	7.08	6.64	6.99	6.89	6.00	5.99	6.01	6.62	7.77	8.36
	HDD50	1724	444	380	193	77	13	0	0	0	0	40	226	351
	HDD65	5075	903	794	631	443	217	90	23	30	120	366	655	803
	CDD50	2595	6	6	27	85	278	444	609	591	372	144	21	12
	CDD65	471	0	0	0	1	17	83	168	156	42	4	0	0
	CDH74	3453	0	0	2	10	135	601	1332	1139	215	19	0	0
	CDH80	987	0	0	0	0	9	153	436	352	34	3	0	0
Wind	WSAvg	7.0	7.9	7.1	8.2	7.4	7.0	7.1	6.5	6.4	6.5	6.5	6.5	6.7
Precipitation	PrecAvg	37.9	2.5	2.1	2.5	3.1	4.3	4.8	3.7	4.1	2.8	2.6	2.9	2.4
	PrecMax	48.1	4.0	4.8	5.0	7.1	9.6	8.6	6.4	11.4	6.5	6.8	5.7	6.6
	PrecMin	26.9	0.5	0.7	0.8	1.2	1.6	1.0	1.4	1.2	0.7	0.5	0.9	0.1
	PrecStd	5.4	1.0	1.0	1.1	1.6	1.9	1.9	1.5	2.2	1.7	1.7	1.2	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	62.6	72.0	75.7	81.2	88.7	93.2	93.1	84.2	76.0	63.9	62.5
		MCWB	49.3	50.6	54.4	56.6	61.4	69.0	70.2	68.8	64.6	61.2	53.6	50.6
	2%	DB	54.5	57.0	62.8	70.0	78.0	84.4	88.2	88.0	79.6	69.7	59.8	58.2
		MCWB	47.2	48.2	50.4	53.2	60.0	67.0	68.5	67.2	64.3	58.1	52.7	49.8
	5%	DB	51.5	52.3	58.3	65.8	74.5	80.6	84.9	83.6	76.0	65.4	56.5	54.5
		MCWB	45.8	46.1	48.3	52.1	59.0	65.2	67.4	66.2	63.2	56.8	51.1	48.8
	10%	DB	48.8	49.2	55.2	61.8	70.9	77.1	81.5	80.2	72.9	62.1	53.7	51.1
		MCWB	44.0	44.4	47.0	49.9	57.9	63.5	66.5	65.1	62.0	55.6	49.4	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	51.2	54.9	57.4	64.6	70.7	72.3	70.6	67.8	62.2	55.7	54.2
		MCDB	57.5	60.9	68.7	71.2	76.2	83.9	85.9	86.4	78.8	71.9	61.2	57.8
	2%	WB	48.2	49.0	51.8	55.2	62.4	68.3	70.3	68.8	65.8	60.2	53.7	50.9
		MCDB	52.9	55.6	61.2	67.6	72.9	81.3	84.3	83.3	75.9	67.8	58.2	56.3
	5%	WB	46.5	46.7	49.6	53.3	60.3	66.5	68.7	67.5	64.3	58.0	51.7	48.7
		MCDB	51.2	51.4	56.4	62.2	70.5	78.3	82.2	80.7	73.6	63.8	55.8	53.9
	10%	WB	44.0	44.7	47.8	51.6	58.8	64.5	67.2	66.2	62.8	56.1	49.3	46.2
		MCDB	48.3	49.2	53.9	60.4	68.7	75.3	79.4	77.8	71.2	61.0	53.5	51.5
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.8	13.0	15.0	16.1	15.9	17.3	17.5	15.7	12.5	10.0	8.5
		MCDBR	11.5	15.9	18.1	21.9	22.0	21.2	22.8	22.9	20.3	17.7	15.0	13.0
		MCWBR	8.2	10.1	9.9	10.8	10.7	9.7	10.3	9.5	9.3	9.4	9.6	9.1
	5% WB	MCDBR	10.8	14.1	15.2	19.0	18.7	20.0	20.5	20.4	18.3	15.6	13.2	12.3
		MCWBR	8.5	9.8	9.2	10.1	10.1	9.5	9.8	9.1	9.2	8.9	9.6	9.4
Clear-Sky Solar Irradiance	taub	0.313	0.338	0.372	0.408	0.418	0.424	0.422	0.417	0.392	0.369	0.335	0.310	
	taud	2.440	2.356	2.304	2.230	2.248	2.271	2.270	2.301	2.347	2.399	2.446	2.441	
	Ebn at Noon	241	258	266	267	268	267	266	261	257	244	230	227	
	Edh at Noon	21	28	35	41	42	41	41	38	33	27	21	19	
All-Sky Solar Radiation	RadAvg	322	570	958	1365	1599	1854	1750	1524	1151	705	368	259	
	RadStd	44	93	144	196	168	161	162	151	116	65	43	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (98 neighbors)	N/A	N/A	N/A	N/A	+0.70	+0.56	N/A	N/A	+88	+31	

Nomenclature: See separate page